

Orion Online School System — Comprehensive User Manual

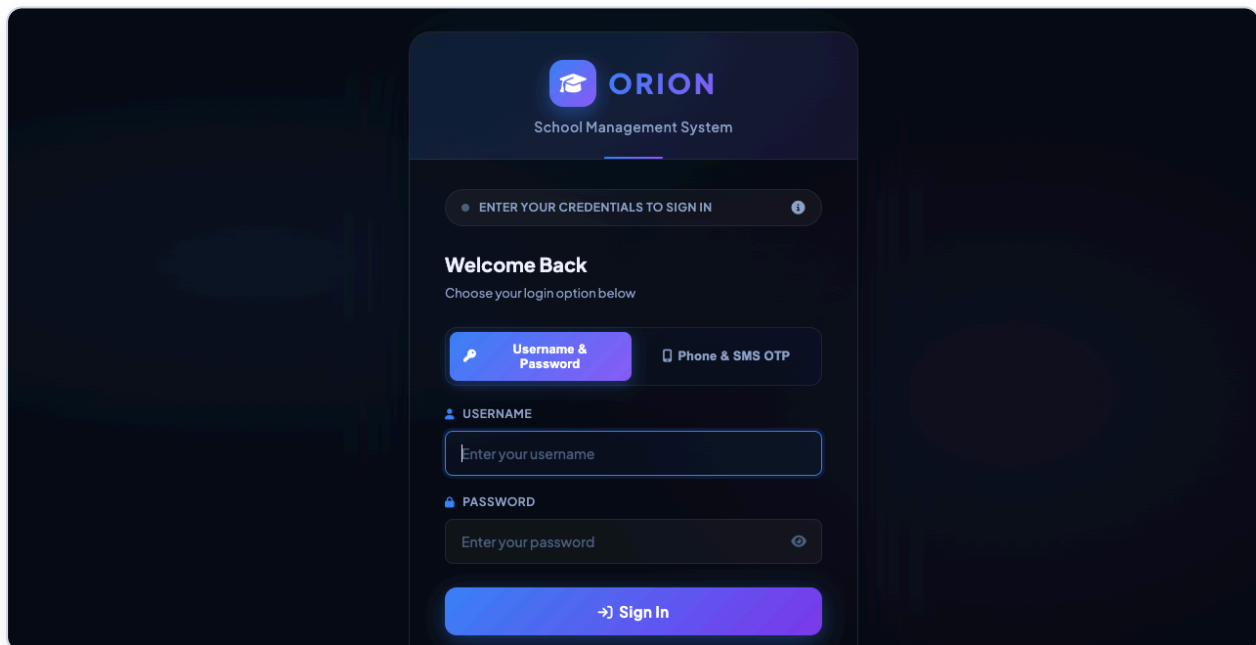
Welcome to the **Orion Online School Management System**. This user manual provides step-by-step instructions for all user roles (Headteachers, Administrators, Teachers, Accountants, Parents, Students, and System Administrators).

1. Getting Started & Authentication

1.1 Accessing the Portal

Open any modern web browser (Google Chrome, Microsoft Edge, Safari, or Mozilla Firefox) and navigate to your school's portal URL:

👉 `https://orion-online-school-system-new.vercel.app` (or your school's dedicated domain).



Login Screen Portal

1.2 Sign In Methods

1. **Standard Username & Password Login:**
2. Enter your assigned **Username** (e.g. `headteacher` , `teacher_kofi`) and **Password**.
3. Click **Sign In to Workspace**.
4. **Phone Number SMS OTP Login:**
5. Click **Request SMS Login OTP**.

6. Enter your registered mobile phone number (e.g. `+233 24 123 4567`).
7. You will receive a 6-digit OTP code on your mobile phone via SMS.
8. Enter the 6-digit code to log in securely without typing your password.

1.3 Progressive Web App (PWA) & Mobile Installation

Orion is fully equipped as a Progressive Web App (PWA). Teachers, administrators, accountants, and parents can install the app directly onto their smartphone, tablet, or desktop home screen without downloading from app stores.

Installing on iOS (iPhone & iPad)

1. Open **Safari** on your iOS device and navigate to `https://orion-online-school-system-new.vercel.app` .
2. Tap the **Share** button (the `[↑]` box icon in the Safari bottom toolbar).
3. Scroll down the menu options and select **Add to Home Screen**.
4. Confirm the app name (**Orion SMS**) and tap **Add** in the top-right corner.
5. The **Orion SMS** icon will appear on your Home Screen, launching in full-screen standalone mode without browser bars.

Installing on Android & Chrome Mobile

1. Open **Google Chrome** on your Android device and navigate to the portal.
2. An **"Install Orion SMS"** banner will slide up at the bottom of the screen. Tap **Install**.
3. If the banner does not appear, tap the three dots `:` menu in the top right corner of Chrome and select **Install app** or **Add to Home screen**.
4. The app icon is added to your home screen and app drawer.

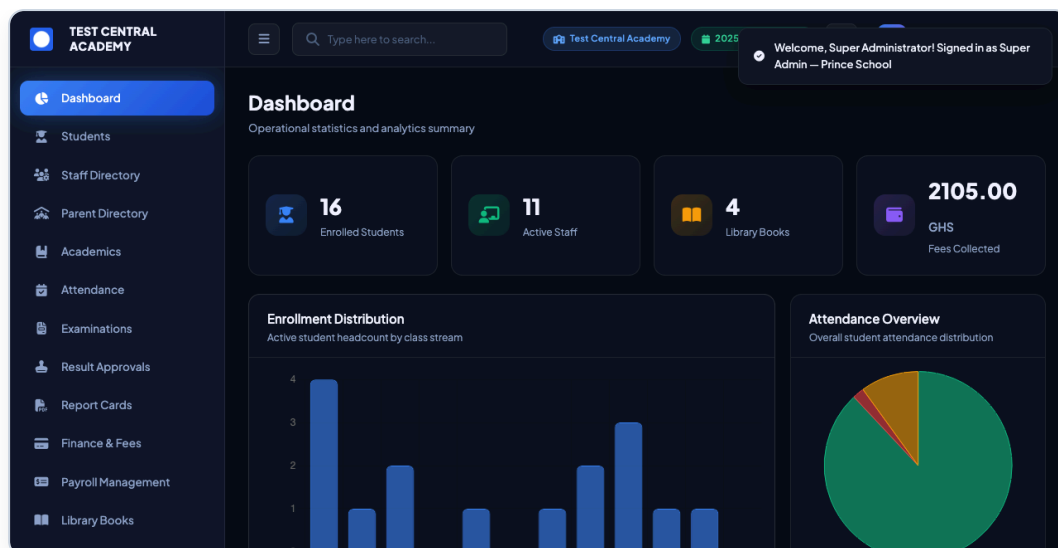
Installing on Desktop (Windows / macOS / ChromeOS)

1. Open the portal in **Google Chrome** or **Microsoft Edge**.
2. Click the **Install Icon** (`⊕` or monitor icon) located on the right side of the browser address bar.
3. Click **Install**. Orion will open in its own dedicated, chromeless window.

2. Dashboard Overview & Workspace Navigation

2.1 Workspace Dashboard

Upon logging in, users are presented with a role-based Glassmorphic dashboard displaying active student metrics, staff counts, library books, termly fee collections, and real-time class distribution analytics charts.



Dashboard Overview

2.2 Real-Time Notification Badges & Live Header Bell

The system features a **Server-Sent Events (SSE) Real-Time Notification Engine** that broadcasts live updates to connected users without needing manual page refreshes.

1. **Header Notification Bell:**
2. Located in the top header action bar.
3. Features a glowing red count badge (`notif-badge-global`) summarizing unread events.
4. Clicking the bell opens the **Live Notification Panel**, displaying:
5. **Unread Parent Messages**
6. **Pending Fee Bills & Unpaid Accounts**
7. **New Announcements & Notices (Past 7 days)**
8. The green pulsing dot (**Live Dot**) confirms an active real-time SSE stream with the backend server.
9. **Sidebar Navigation Badges:**
10. **Communication Badge:** Displays live unread messages sent by parents.
11. **Finance & Fees Badge:** Displays live count of unpaid student bills.

3. Headteacher & Administrator Guide

3.1 Updating School Profile, Logo & Digital Signature

1. Navigate to **Settings** in the left sidebar menu.
2. Under the **School Profile & Branding** tab:

3. Enter **School Name, School Motto, Tagline, Phone Number, Email, and GPS Address.**
4. **Upload School Logo:** Click *Choose File* under School Logo. Select a PNG or JPG logo image. The logo updates immediately and is saved securely to the database.
5. **Upload Headteacher Signature:** Click *Choose File* under Headteacher / Authorization Signature. Upload a transparent PNG image of the headteacher's signature. This signature will automatically watermark official terminal report cards and transcripts.
6. Click **Save School Profile.**

TEST CENTRAL ACADEMY

Parent Directory
Academics
Attendance
Examinations
Result Approvals
Report Cards
Finance & Fees
Payroll Management
Library Books
Inventory
Timetables
Communication
Configurations

System configurations
General profile parameters, academic calendar, auto-backups, and grading rules

School Profile Info | Academic Years & Terms | Data Backups | Grading Scales | Payment Gateway

SCHOOL NAME (CONFIGURED BY SYSTEM ADMIN)
Test Central Academy

SCHOOL MOTTO
Test Motto

SCHOOL TAGLINE / ID PREFIX
PS
Used as student ID prefix (e.g. ORION-26-XXXX)

SCHOOL PHONE
+233 24 111 2222

SCHOOL EMAIL
admin@testcentral.edu.gh

PHYSICAL ADDRESS
123 Innovation Drive

GPS ADDRESS
GA-990-1122

SCHOOL LOGO UPLOAD
 Choose File No file chosen
Displays on Header, Report Cards & Exported PDFs

Save Profile Details

School Profile & Settings Panel

3.2 Headteacher Profile & Authorization Signature Modal

Headteachers and authorized administrative staff can also manage their account settings and signature directly from the **My Profile & Account Settings** modal.

My Profile & Account Settings

Personal Details | Security & Password

USERNAME
Assigned Role

FIRST NAME
Last Name

EMAIL ADDRESS
e.g. user@school.edu

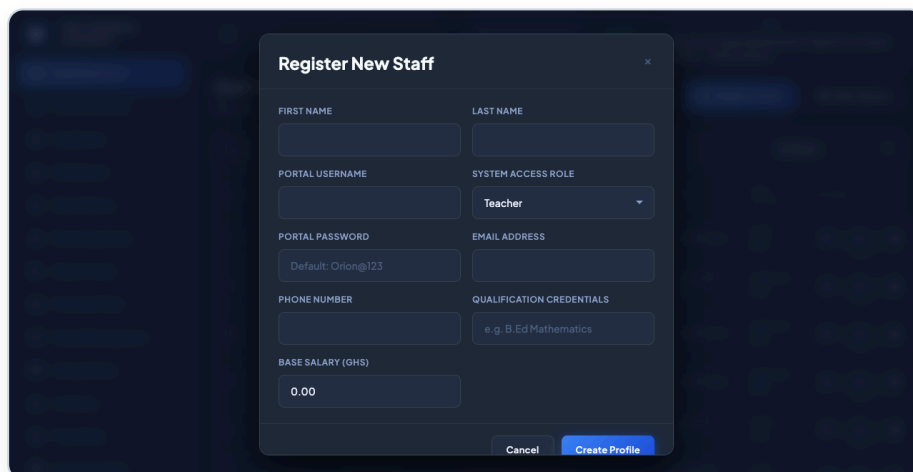
PHONE NUMBER
e.g. +233241234567

HEADTEACHER / AUTHORIZATION SIGNATURE
No signature | Choose File No file chosen
Upload transparent PNG signature image (rendered on approved report cards & certificates)

Headteacher Signature Modal

3.3 Registering Staff Members & Automated SMS Credentials

1. Navigate to **Staff Directory** in the left sidebar.
2. Click **+ Register Staff**.
3. Fill in the staff member's details:
4. **First Name & Last Name** (e.g. *Kwame Appiah*)
5. **Assigned Role:** Select *Teacher*, *Headteacher*, *Accountant*, *Librarian*, or *IT Admin*.
6. **Username:** Create a unique username (e.g. *kwame.appiah*).
7. **Mobile Phone Number:** Enter their valid 10-digit mobile phone number (e.g. *0241112233*).
8. **Initial Password:** Enter a password (or leave blank to assign default *Orion@123*).
9. Click **Register Staff Profile**.
10. **Automated SMS Action:** The system automatically dispatches an SMS to the new staff member's mobile phone containing their username, initial password, and portal link.



The image shows a 'Register New Staff' modal form. It contains the following fields and options:

- FIRST NAME**: Text input field.
- LAST NAME**: Text input field.
- PORTAL USERNAME**: Text input field.
- SYSTEM ACCESS ROLE**: Dropdown menu with 'Teacher' selected.
- PORTAL PASSWORD**: Text input field with a default value 'Default: Orion@123'.
- EMAIL ADDRESS**: Text input field.
- PHONE NUMBER**: Text input field.
- QUALIFICATION CREDENTIALS**: Text input field with a placeholder 'e.g. B.Ed Mathematics'.
- BASE SALARY (GHS)**: Text input field with a value '0.00'.

At the bottom right of the modal are two buttons: 'Cancel' and 'Create Profile'.

Register Staff Modal

3.4 Approving Class Terminal Reports

1. Navigate to **Exams & Reports -> Score Approvals**.
2. Review submitted class marksheets submitted by subject teachers.
3. Verify Class Scores (30%), Exam Scores (70%), and calculated Total Scores.
4. Click **Approve Assessment Roster**. Approved scores are immediately locked and rendered onto official student terminal report cards.

3.5 Batch Printing Student Terminal Report Cards

1. Navigate to **Exams & Reports -> Report Cards**.

2. Select the **Academic Term** and **Class** (e.g., *Basic 7 - Alpha*).
 3. Click **Generate Batch PDF Report Cards**.
 4. The system compiles official Report Cards containing:
 5. School Logo Header & School Contacts.
 6. Student Rank / Position in Class.
 7. Subject Grade Breakdown, Totals & Remarks.
 8. Embedded Headteacher Signature Watermark.
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4. Teacher Guide

4.1 Marking Daily Student Attendance

1. Navigate to **Attendance** in the left sidebar.
2. Select your assigned **Class** and the **Date**.
3. Click **Present**, **Absent**, or **Late** for each student.
4. Click **Save Attendance Record**.
5. *Automatic Parent Alert*: Parents of absent students automatically receive an instant SMS attendance notification.

4.2 Recording Continuous Assessment & Exam Scores

1. Navigate to **Exams & Reports -> Marks Entry**.
2. Select the **Class**, **Subject** (e.g., *Mathematics*), and **Academic Term**.
3. Enter scores into the roster:
4. **Class Score**: Max 30 marks (Continuous Assessment / Homework / Tests).
5. **Exam Score**: Max 70 marks (Terminal Examination).
6. Total Score, GES Grade (A, B, C, D, F), and Grade Remarks calculate automatically.
7. Click **Submit Marks to Headteacher for Approval**.

4.3 Offline Attendance & Mark Entry (Service Worker Sync)

Teachers working in remote areas or classrooms with unstable Wi-Fi/cellular connection can record attendance and enter marks completely offline.

1. **Automatic Offline Detection**:

2. If internet connectivity drops, an amber banner appears at the top of the app: *"You're offline — data is cached. Queued actions will sync when reconnected."*
 3. The Service Worker (`sw.js`) automatically serves cached student rosters and application interfaces.
 4. **Offline Data Queuing:**
 5. When you click **Save Attendance** or **Submit Marks** while offline, the system safely stores your submission in the device's **IndexedDB Sync Queue**.
 6. You will see a toast notification confirming: *"Saved offline. Will sync when connection restores."*
 7. **Background Synchronization:**
 8. As soon as your device reconnects to Wi-Fi or mobile data, the Service Worker automatically executes a **Background Sync**, transmitting all queued attendance logs and mark entries to the server.
 9. A success alert notifies you: *"Offline actions synced successfully"*.
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5. Accountant & Finance Guide

5.1 Setting Up Termly Fee Structures

1. Navigate to **Finance & Billing** -> **Fee Structures**.
2. Click **+ Add Fee Item**.
3. Specify Fee Title, Class Level, and Amount (GHS).
4. Click **Save Fee Structure**.

5.2 Recording Student Fee Payments & SMS Receipts

1. Navigate to **Finance & Billing** -> **Collect Payments**.
2. Search for the student by **Name** or **Student ID** (e.g., `ORION-26-A1B2`).
3. Enter Payment Amount, Payment Method (Cash/MoMo/Bank), and Reference Number.
4. Click **Process Payment & Generate Receipt**.
5. **Instant SMS Action:** An SMS receipt is instantly sent to the parent's mobile phone

detailing payment amount and remaining balance.

5.3 Monthly Staff Payroll & Payslip Downloads

1. Navigate to **Finance & Billing** -> **Staff Payroll**.
 2. Select the **Pay Period** (e.g., *July 2026*).
 3. Click **Generate Monthly Payroll**. Review base salaries, allowances, and deductions.
 4. Click **Process Payouts**. Download official PDF Payslips for staff distribution.
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6. Parent & Student Guide

6.1 Accessing the Parent Portal

1. Open `https://orion-online-school-system-new.vercel.app` (or launch the installed Orion PWA app).
2. Click **Request SMS Login OTP** or sign in with your registered phone number and PIN.
3. Select your linked ward/child (if you have multiple children enrolled).

6.2 Viewing Terminal Reports & Transcripts

1. On the Parent Dashboard, navigate to **Academic Reports**.
2. Select the **Term** to view terminal report card details online or click **Download PDF Report Card**.

6.3 Checking Fee Balances & Payment History

1. Navigate to **Financial Ledger**.
2. View total billed fees, total payments made, current outstanding balance, and download past digital receipts.

7. System Administrator Guide

7.1 Multi-Branch Provisioning

1. Log in with System Admin credentials (`sysadmin`).
2. Navigate to **System Admin Portal** -> **Branches**.
3. Click **+ Create Branch**.
Specify Branch Name, Branch Code, and Tenant Database filename.

7.2 Managing SMS Gateway Configuration

1. Navigate to **System Admin Portal** -> **SMS Gateway Config**.
2. Select Gateway Provider (*Arkesel, Hubtel, mNotify, or Twilio*).
3. Enter your **Sender ID** (e.g., `ORION`), **API Key**, and **API Secret**.
4. Click **Test SMS Gateway**. The system sends a live test SMS to verify connection.